Cornea, Conjunctiva, and Sclera Anatomy

Images used: 2.2, 2.5, 2.10, 2.1, 2.8, 2.9, 2.12, 2.14

I. Corneal layers

A. Epithelium

1. 5-7 cells thick

2. Surface epithelial cells (squamous cell layer)

    a. Secretes a glycocalyx that adjoins the mucin layer of the tear film.

    b. Has projections on the outer surface that enhance tear stability

3. Tight junctions (zonula occludens) join the lateral cell walls

4. Clinical correlate: fluorescein dye

5. The basal layer of the epithelium connects attaches onto the underlying tissue by hemidesmosomes.

6. Clinical correlate: recurrent corneal erosion

B. Bowman layer

1. Dense sheet of interwoven collagen fibrils

2. Acellular

3. Does not regenerate

C. Stroma

1. About 90% of the corneal thickness

2. Composed of collagen fibrils, keratocytes and extracellular ground substance

3. The very regular arrangement and diameter of the stromal fibrils, as well as the restriction of the distance between fibrils, contribute to stromal transparency.

4. Clinical correlate: corneal opacification

D. Descemet membrane

1. The basement membrane of the corneal endothelium

2. It is produced throughout life and therefore thickens over a lifetime

3. Resistant to damage, but can be regenerated if damaged

4. As Descemet membrane ends at the limbus, it is called Schwalbe line.

E. Endothelium

1. Single layer of flattened hexagonal cells

2. Ionic pumps help maintain hydration of the cornea

3. Endothelial cells to not divide and replicate

II. Corneal innervation

A. Branches of the long and short ciliary nerves.

B. Plays a role in corneal structure and function

III. Corneal nutrition

A. The cornea gets oxygen primary from atmospheric oxygen that is dissolved in the tears.

1. Small amounts from the aqueous and limbal capillaries.

B. It obtains most nutrients from the aqueous.

1. It also gets some from the conjunctival and episcleral vessels in the limbus.

C. Clinical correlate: corneal neovascularization

IV. Scleral anatomy

A. Functions

1. Maintains the shape of the globe
2. Resistance to external forces
3. Attachment for the extraocular muscles

B. Thick, dense connective tissue

C. The sclera is continuous with the corneal stroma at the limbus

D. Scleral spur

1. Circularly oriented collagen fibrils at the inner, anterior aspect of the sclera.
2. Origin of the ciliary body muscle fibers.

E. Lamina cribrosa

1. Scleral tissue that bridges the foramen in with the optic nerve passes.

F. Blood supply

V. Episclera

A. Loose vascularized connective tissue just outer to the sclera

B. Blood supply to sclera and peripheral cornea

VI. Tenon capsule

A. This fibrous sheet of connective tissue outer to the episclera

B. Protects the globe

VII. Conjunctiva

A. Bulbar conjunctiva

B. Fornix

C. Palpebral conjunctiva

VIII. Limbus

A. Corneoscleral junction

B. Boundary: where Bowman layer and Descemet membrane end